

**LISTENING**
#27**Teacher's Guide****SCRIPT: AN UNUSAL SUPERNOVA**

Interviewer: A supernova is the terminal explosion of a dying star, but scientists have discovered a very unusual example. It doesn't seem to follow the normal pattern. So what is the normal pattern? A question for our science reporter Paul Rinkon.

Rinkon: Well, the normal pattern is that when a star is coming to the end of its life, it can go one of either two ways. It can either collapse into a black hole, or it can explode in this massive energetic event called a supernova. And that releases the energy equivalent to 100 million suns over quite a short space of time-about 100 days. What happened in this case is that it continued for a much longer time. And that's something that people haven't seen before. Astronomers hadn't seen this before, and this seems like a very very unusual object.

Interviewer: And when you say a much longer time?

Rinkon: Yeah. So the typical period duration for a supernova is about 100 days. It doesn't go beyond that. But this one continued for 600 days, and it continued releasing energy. There was no let up. So it released probably the amount of energy about 600 million suns, equivalent to that.

Interviewer: And what scientists think this means?

Rinkon: Well, what they think it means is that perhaps this is a very unusual type of sun that got so massive, got so big, and got so hot, that it actually produced antimatter in its core. Now this is the actual, the opposite of matter. And what happens in this case is that the sun gets violently unstable. And instead of exploding in just one big go, releasing all its energy and then it's gone, it's reduced to this tiny neutron star. It kind of, it just let it out slowly.

Interviewer: Now you keep saying that this is a very unusual event. Does this mean that they are going to have to rethink the way that they look at a supernova, or is this just a huge very energetic blip?

Rinkon: I think the universe is so big, and we haven't sampled enough of the sky. And we just haven't seen these stars before. What's also interesting is that they've seen something, they've seen an explosion from this object before, 50 years ago. And they just can't explain that. So this may actually have been instead of 600 days, it may have been going off essentially in its death throes for 60 years or longer and letting out these short explosions without actually destroying itself in one massive supernova as happens with most stars.

Interviewer: That was our science reporter Paul Rinkon.

Answer Key

Comprehension Activity

1. b) It either collapses into a black hole or explodes as a supernova
2. a) The energy of 100 million suns
3. b) 100 days
4. b) It continued for 600 days
5. b) Antimatter
6. It would make the star violently unstable / release energy slowly over time
7. a) A small neutron star
8. a) 60 years or longer
9. Because the universe is huge and we haven't observed many of these unusual stars

Reading Comprehension

1. They produce heavy elements and energy; they help form planets, stars, and matter in the universe.
2. b) Their behavior doesn't match normal patterns
3. It may explode because of interaction with a companion (like a black hole) — not just by a single collapse.
4. b) The explosion becomes more luminous and lasts longer

Idioms

“Death throes” = the final struggles before the end

“Let up” = to lessen or stop (of an action or process)